

Haoyang Jiang

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RESEARCH PROFILE

Ph.D. researcher in **scientific machine learning for physical dynamical systems**, spanning neural operators, physics-guided forecasting, differentiable simulation, inverse problems, and data assimilation. Background in geophysics and numerical dynamics, with Python/PyTorch-based research and reproducible evaluation.

EDUCATION

William & Mary, Williamsburg, VA *Expected 2027*
Ph.D. in Data Science

University of Science and Technology of China, Hefei, China *2023*
M.S. in Astrometry and Celestial Mechanics

China University of Mining and Technology (Beijing), Beijing, China *2020*
B.S. in Geophysics; Minor in Computer Science

SELECTED RESEARCH

- **Neural Operators and Physics-Guided Forecasting** [\[GitHub\]](#): Developed Fredholm-integral methods for PDE boundary-value problems and open-boundary hydrodynamic models with learned boundary context and direction-aware physics refinement; built a shallow-water finite-volume solver for validation.
- **Differentiable Wave-Equation Inverse Modeling** [\[GitHub\]](#): Built acoustic forward modeling and full-waveform inversion in PyTorch/autograd, combining multiscale objectives, learned generative priors, and gradient-based optimization; background in seismic forward modeling and inversion.
- **Data Assimilation and Uncertainty Propagation**: Built assimilation workflows for sparse data and distribution shift; M.S. research quantified orbital-state error growth under atmospheric-density uncertainty.
- **Flow Maps and Generative Modeling**: Investigate continuous-time flow maps, composition consistency, and flow matching for generative modeling.
- **Neural Differential and Integral Equations** [\[Demos\]](#): Implement Neural ODE, learned-delay DDE, and NIE models with differentiable solvers and adjoints; evaluate long-horizon sensor prediction.

SELECTED PUBLICATIONS

1. H. Jiang, Z. Wang, S. Gao, Y. J. Zhang, X. Zhu, Y. He. *Physics-Refined Spatiotemporal Forecasting on Open-Boundary Hydrologic Graphs*. **IEEE ICDM**, 2026. **Selected among the best-ranked papers; invited for KAIS extension.**
2. H. Jiang, B. Qu, X. Zhu, J. Tan, Y. He. *Boundary-Consistent Graph Neural Networks for Topological Flux Prediction*. **TMLR**, 2026.
3. H. Jiang, Y. Qu. *Fredholm Integral Equations Neural Operator (FIE-NO) for Boundary Value Problems*. **Machine Learning: Engineering**, 2026.
4. H. Jiang, J. Wang, Y. He, et al. *Topology-aware Neural Flux Prediction Guided by Physics*. **ICML**, 2025.
5. H. Jiang, M. Zhang, et al. *Orbital error propagation considering atmospheric density uncertainty*. **Advances in Space Research**, 2023.

ENGINEERING EXPERIENCE

ORIGIN SPACE — Assistant Engineer Intern *Feb. 2022 – Jun. 2022*

- Reproduced and benchmarked machine-learning and numerical algorithms for aerospace engineering R&D, evaluating accuracy, robustness, and computational cost; improved research prototypes for practical deployment.

TECHNICAL SKILLS

Scientific ML: Neural operators, physics-informed ML, GNNs, flow/generative models, Neural ODE/DDE/NIE.

Numerical Methods: PDE/ODE solvers, finite volumes, inverse problems, optimization, data assimilation.

Programming: Python, PyTorch/autograd, PyG, NumPy/SciPy, MATLAB.